

Effects of Transfer Functions and Body Parts on Body-Centric Locomotion in Virtual Reality

BoYu Gao^{ID}, Zijun Mai, Huawei Tu^{ID}, and Henry Been-Lirn Duh^{ID}

Abstract—Body-centric locomotion allows users to control both movement speed and direction with body parts (e.g., head tilt, arm swing or torso lean) to navigate in virtual reality (VR). However, there is little research to systematically investigate the effects of body parts for speed and direction control on virtual locomotion by taking in account different transfer functions (L: linear function, P: power function, and CL: piecewise function with constant and linear function). Therefore, we conducted an experiment to evaluate the combinational effects of the three factors (body parts for direction control, body parts for speed control, and transfer functions) on virtual locomotion. Results showed that (1) the head outperformed the torso for movement direction control in task completion time and environmental collisions; (2) Arm-based speed control led to shorter traveled distances than both head and knee. Head-based speed control had fewer environmental collisions than knee; (3) Body-centric locomotion with CL function was faster but less accurate than both L and P functions. Task time significantly decreased from P, L to CL functions, while traveled distance and overshoot significantly increased from P, L to CL functions. L function was rated with the highest score of *USE-S*, *-pragmatic* and *-hedonic*; (4) Transfer function had a significant main effect on motion sickness: the participants felt more headache and nausea when performing locomotion with CL function. Our results provide implications for body-centric locomotion design in VR applications.

Index Terms—Human-centered computing, human computer interaction (HCI), interaction paradigms, virtual reality

1 INTRODUCTION

As a common task in Virtual Reality (VR), locomotion can be executed with different methods such as using controllers [5], [6] or body parts [18], [40], [59]. Body-centric locomotion is an important way of navigating in VR [17], [37], [51]. In principle, such a way changes virtual viewpoint (i.e., user viewpoint in VR) according to body movements. There are two categories of body-centric locomotion techniques: gait-enabled and gait-free types [39]. The former one requires users to walk in physical environments to cause virtual movements in VR, while the latter one relies on body-part movements while standing in place (see Fig. 1 as an example).

We focus on the gait-free locomotion due to the following advantages. First, compared to the gait-enabled locomotion, the gait-free locomotion could be easily adapted in virtual scenes with a limited physical space as it does not need a dedicated room space [6], [16], [40], [53]. Second, the gait-

free locomotion generally depends on body parts to control virtual movements without the need to carry or hold auxiliary input devices such as controllers, hence creating always-available interaction and allowing users to transition between locomotion and other tasks efficiently and naturally [14], [15], [36], [56]. Third, the gait-free locomotion can support concurrent task execution with one body part assigned to locomotion tasks and other body parts for different tasks [17], [47], [51]. Thanks to these strengths, previous studies have proposed many techniques based on the gait-free locomotion, including using the head, arm, torso, and leg (e.g., [18], [20], [33], [40], [51], [59]).

Movement displacement (speed) and direction are two important determinants of the locus of the gait-free locomotion. Users usually need to control movement displacement (speed) and direction simultaneously in locomotion tasks [14], [15], [40], [52]. Movement displacement (speed) control generally relies on transfer functions to transform body motion into their virtual movement. Existing studies have applied different transfer functions to locomotion design with a certain body part, such as torso-based locomotion using a linear function [18] or a piecewise function with linear and power functions [34], hand-based locomotion using a power function [25] or a piecewise function with linear and power functions [26], [34]. Our previous work first explored the effects of transfer functions on movement speed control with different body parts for virtual locomotion [4], and the results showed locomotion performance varied in different combinations of body parts and transfer functions. While providing insights into body-centric locomotion, we did not focus on how to control movement direction with body parts.

Direction control refers to human ability to maintain their body orientation relative to the surrounding environment [7].

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Manuscript received 19 Nov. 2021; revised 9 Apr. 2022; accepted 18 Apr. 2022. Date of publication 21 Apr. 2022; date of current version 30 June 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 61902147, in part by Guangdong Province under Grant 2021A1515012629, in part by Guangzhou Applied and Basic Applied Foundation under Grant 202102021131.

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by Jinan University.

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Recommended for acceptance by F. Steinicke.

Digital Object Identifier no. 10.1109/TVCG.2022.3169222



Fig. 1. Illustration of head-tilt-based locomotion. (a) user's initial viewpoint in VR without head tilt. (b) Change of the user's viewpoint in VR along with head tilt.

It relies heavily on perceived visual information and whole-body information while moving in VR [24]. Previous studies showed the difficulty of maintaining direction in VR [7], [8], [50]. In particular, maintaining direction while manipulating locomotion speed could be a challenge.

To extend the generalizability of our previous work [4], we further investigated the effects of body parts and transfer functions on body-centric locomotion with movement direction control included. We aimed to understand how each body part cooperates to generate the efficient or accurate or comfortable body-centric locomotion experience. We conducted an experiment to address the above needs. Based on previous results [4], we selected three body parts for movement speed control (head, arm and knee) with three transfer functions, and two body parts for movement direction control (head and torso). The three transfer functions were a linear function [5], [6], [18], a power function [15], [25], [40], [59], and a piecewise function composed of linear and constant functions [17], [37]. The three body parts for speed control and the two body parts for direction control form six combinations: head + head [37], [47], [56], head + arm [14], [15], [40], [52], head + knee [46], torso + head [18], torso + arm [18], and torso + knee [20], [28], [38], [51].

The contribution of this paper can be summarized in three parts. First, we thoroughly reviewed the existing work on body-centric locomotion in VR, providing rationales to select body parts for direction control and speed control, transfer functions, and task types. Second, we investigated the combinational effects of body parts and transfer functions on body-centric locomotion. With the inclusion of body parts for direction control, we obtained new findings for body-centric locomotion in VR, compared with previous studies [4], [17], [18]. For example, the combination of head (direction control) and arm (speed control) with L function could achieve the best locomotion performance while keeping comfortable experience, and the combination of torso (direction control) and knee (speed control) with P function could lead to hands-free locomotion while keeping slow movement. Third, we presented design implications based on the findings of this experiment. Designers can select a proper combination of body parts with a corresponding transfer function according to locomotion requirements (e.g., efficiency, accuracy, comfortable experience, concurrent tasks or hands-free locomotion) for VR applications (e.g., VR games, VR museum and VR education).

2 RELATED WORK

This section summarizes the existing work on using body parts and transfer functions for virtual locomotion, and locomotion tasks in VR.

2.1 Body-Centric Virtual Locomotion

The body parts used in previous studies for direction and speed control of virtual locomotion mainly include the head, torso, hand/arm, knee and foot.

Head-based locomotion allows users to tilt or yaw the head to control the movement of the user's viewpoint in VR. Movement direction control with the head has been examined in previous studies. Pfeiffer *et al.* [57] manipulated movement direction by yawing the head recognized by inertial sensors. Tregillus *et al.* [55] also controlled movement direction by yawing the head and movement speed by measuring step frequency with inertial measurement units in smartphones. McCullough *et al.* [40] used the head to control the direction of the user's viewpoint and adjusted the speed of locomotion with arms-swinging by a wearable armband. Zielasko *et al.* [17] evaluated five different locomotion methods, including a head-directed control with two accelerator pedal metaphors.

Using the head to control both movement speed and direction is a common way in hands-free VR applications. Tregillus *et al.* [56] proposed a hands-free VR navigation approach with head tilt, freeing two hands for other tasks. Terziman *et al.* [37] proposed *shake your head*, which utilized head tilting to rotate the viewpoint and head motions to control forward and jump in VR.

Torso-based Locomotion is to lean it forward to adjust user's viewpoint in VR [3], [17], [20], [21], [22], [51]. Such a method has been verified to positively affect self-motion perception [21]. Users can adopt this torso-directed manner to control movement direction by yawing or leaning their torso. Yan *et al.* [38] relied on the movement of users' calf to adjust locomotion speed and yawed torso to control direction. Wendt *et al.* [28] also used the combination of knee and torso to control moving forward, but they utilized the frequency of knee instead of lifting speed. Kim *et al.* [31] measured stepping frequency by a sensor and controlled the direction by yawing torso. Similar to the study of Kim *et al.* [31], Feasel *et al.* [29] employed the vertical heel speed of feet to control moving speed. In addition, Guy *et al.* [20] controlled movement direction by yawing or leaning torso.

Leaning the torso is a common way to adjust the speed of virtual locomotion. Zielasko *et al.* [17] found that torso-based locomotion outperformed both head-based and gamepad-based locomotion methods. Buttussi *et al.* [22] also revealed that torso-based locomotion was better than joystick-based, but worse than teleport in terms of locomotion efficiency. Ganapathi *et al.* [49] studied various body gestures (i.e., leaning, tapping and pointing), and identified the leaning gesture suitable to travel tasks while enabling object selection and manipulation simultaneously. However, according to the study of Gao *et al.* [4], compared to the head, arm and knee, the torso was not less preferred for speed control, probably due to the tiredness of performing locomotion with torso while standing during the experiment.

Hand/arm-based locomotion commonly requires users to swing their arm to control the speed of the user's viewpoint in VR [15], [40], [52], [59]. McCullough *et al.* [40] proposed the use of *arm swing* (tracked by a MYO Armband) to perform locomotion and demonstrated its efficacy in virtual locomotion. In addition, Pai *et al.* [59] implemented *arm swinging* with HTC controllers and adjusted the speed by

the amplitude of swinging, and found that it required less efforts and allowed prolong uses for navigation in AR/VR. However, Wilson *et al.* [52] found that *arm swing* was not good for spatial orientation tasks, resulting in more turning errors than both real walking and walking-in-place methods in VR. Some studies used fingers to control the direction of virtual locomotion. For example, Kim *et al.* [32] adopted two fingers sliding alternately to control the direction of the user's viewpoint.

Knee-based locomotion methods are mainly employed to control the movement speed of user's virtual viewpoint by bending or yawing the knee [20], [46], [51]. Nilsson *et al.* [46] presented a *wipping* method for virtual navigation by bending the knee while keeping the upper leg relatively steady. Guy *et al.* [20] proposed *LazyNav*, which uses body parts (e.g., knee, shoulder and foot) for 3D ground navigation in large displays, enabling a hands-free mid-air ground navigation control. Punpongsanon *et al.* [51] adopted *LazyNav* in VR interaction, so that users can navigate virtual scenes with the knee or the shoulder while being able to perform other tasks with the hands and the head. In addition, Templeman *et al.* [30] transformed the displacement or the angle of knee during stepping to control the direction and the speed of virtual viewpoint.

Foot-based locomotion methods commonly adopt the combination of marching and yawing the feet for speed and direction control [39], [35], [58]. However, we did not adopt the foot mainly because foot-based methods are commonly used to simulate realistic walking or perform teleportation (e.g., [12], [33]). As a representative foot-based locomotion style, walking-in-place (WIP) techniques have been well researched in previous studies (e.g., [39], [48], [58]). They usually map footstep frequency to walking speed in virtual environments [48]. However, such a mapping mechanism (i.e., transfer function) was seldom applied to body parts examined in our study for virtual locomotion (i.e., the head, arm and knee). As the purpose of our study is to investigate the interrelationship between body parts and transfer functions in the context of virtual locomotion, our experiment design needs to include body parts that were commonly used in previous locomotion studies, and also transfer functions that can be applicable across the selected body parts. Therefore, we did not evaluate the performance of the foot in our study.

In addition, only a few works made use of the hip or shoulder to perform locomotion [20], [36], [51]. As they are not common body parts for virtual locomotion, we did not take them into account in our experiment design.

Based on our review, we selected three common body parts (i.e., head, arm, knee) for speed control and two common body parts (e.g., head [37], [47], [56] and torso [18]) for direction control in virtual locomotion for this work.

2.2 Transfer Functions for Body-Centric Locomotion

Transfer functions are essential to design body-centric locomotion. The purpose of transfer functions is to update the speed or displacement of the user's viewpoint in VR according to body motion, leading to a virtual walking experience. Below we summarize typical transfer functions for body-centric virtual locomotion from existing studies [4].

Linear functions have been used in many locomotion studies [5], [6], [17], [18], [36]. Linear functions have two main types. One linearly transfers the angle of body motion to the displacement of the user's viewpoint in VR [5], [6], [18]. (1) in Table 1). The other is linearly mapping the step height or step speed to the speed of the user's viewpoint [17], [36] (2) in Table 1). The steps were tracked by depth cameras [36] or detected by using the inertial measurement unit of a smartphone carried in pants pocket [17].

Power functions in previous studies [14], [15], [25], [40], [52], [59] can be classified into two categories. One is the cubic function with the displacement of body movement as the variable and the displacement of the user's viewpoint in VR as the function value [25] (3) in Table 1). The other is the quadratic function with the angle of body movement as the variable and the displacement of the user's viewpoint in VR as the corresponding function value [14], [15], [40], [52], [59] (4) in Table 1).

Piecewise functions used in previous studies vary in forms and mapping types. The piecewise function with constant and linear functions maps the displacement of a tracked body part to the movement speed of the user's viewpoint in VR (5) in Table 1). The piecewise function comprising linear and power functions transfers the displacement of a tracked body part to the displacement of the user's viewpoint in VR (6) in Table 1). The piecewise function composed of constant and power functions transfers the angle of the movement of body parts to the movement speed of the user's viewpoint in VR (7) in Table 1).

Most of the above transfer functions were examined in the context of locomotion with one or two body parts. This study aims to compare their performance across different body parts for speed and direction control, so as to achieve more generalized conclusions by considering varying forms and mapping mechanisms of transfer functions for virtual locomotion.

2.3 Experiment Task Design for Body-Centric Locomotion

We review experiment task design for body-centric locomotion from two categories: movement path and task type.

Movement Path. There were three main movement path types for body-centric locomotion in previous studies. The first is a single straight path with some targets [1], [4] (Table 2:1a). The second is a curved path including some targets [1], [18], [20], [46], [51] (Table 2:1b). The third is a path with line segments connected by a number of targets [2], [3], [6], [9], [11], [19], [22], [34], [36], [37], [40] (Table 2:1c).

Task Type. Previous studies mainly adopted two task types for body-centric locomotion. One is locomotion tasks without secondary tasks. Such a task type has two main styles in existing studies. One is following the assigned path to reach a target (Table 2:2a). Usually there is only one predefined curved or straight path in the virtual scene. In previous studies [46], [51], participants were asked to follow a predefined path indicated by green color. If participants were out of the path, feedback would be given by turning the path from green to red. The other style is to ask a participant to reach multiple targets one by one [3], [11], [19], [22],

TABLE 1
A Summary of Body-Centric Locomotion With Relevant Transfer Functions

References	Direction Control	Speed Control	Transfer Function	Mapping Mechanism
Zielasko et al. [18] Sarupuri et al. [5], [6]	Head yawing, torso yawing	Torso leaning, hand leaning	Linear Function (L): $v_v = k \times \theta \quad (1)$	Angle to speed mapping: the leaning angle of torso (θ) is mapped to the speed of the user's viewpoint in VR (v_v) with a scale factor k .
Bruno et al. [36] Zielasko et al. [17]	Hip rotating	Foot lifting	Linear Function: $v_v = k \times d_r \quad (2)$	Displacement to speed mapping: the displacement (d_r) of step lift and lower is linearly mapped to the speed of the user's viewpoint (v_v) with a factor k in VR.
Cirio et al. [25]	Hand pushing	Hand pushing	Power Function (P): $v_v = k \times d_r^3 \quad (3)$	Displacement to speed mapping: the displacement of hand movement (d_r) is mapped to the speed of the user's viewpoint in VR (v_v) with a scale factor k .
McCullough et al. [40] Pai et al. [59] Calandra et al. [14], [15] Wilson et al. [52]	Head yawing	Arm swinging	Power Function: $d_v = k \times \theta^2 \quad (4)$	Angle to displacement mapping: the angle of arm swing (θ) is mapped to the displacement of the user's viewpoint in VR (d_v) with a scale factor k .
Terziman et al. [37] Zielasko et al. [17]	Head tilting	Lateral head motions	Piecewise Function with Constant and Linear Functions (CL): $v_r = \frac{\min(\frac{d_r}{t} , 0.5)}{0.5}$ $v_v = \begin{cases} 0 & \text{if } v_r < 0.05 \\ k \times v_r & \text{if otherwise} \end{cases} \quad (5)$	Displacement to speed mapping: 0.05 and 0.5 are two threshold values. d_r means the head movement displacement. t represents time. v_r represents the normalized value ([0, 1]) of head movement speed. v_r is mapped to the speed of the user's viewpoint in VR (v_v) with a scale factor k .
Poupyrev et al. [26] Wang et al. [34] Bowman et al. [13]	Hand pushing	Hand pushing	Piecewise Function with Linear and Power Functions: $d_v = \begin{cases} d_r & \text{if } d_r < a \\ d_r + k \times (d_r - a)^2 & \text{if otherwise} \end{cases} \quad (6)$	Displacement to displacement mapping: the displacement of body movement (d_r) is mapped to the displacement of the virtual hand in VR (d_v). k is a scale factor. a is two thirds of the length of user's arm.
Guy et al. [20] Punpongsanon et al. [51]	Hip rotating , torso leaning, shoulder rotating	Foot stepping, Knee bending, Torso leaning	Piecewise Function with Constant and Power Functions (CP): $v_v = \begin{cases} 0 & \text{if } \theta \in [0, a] \\ k \times (\frac{\theta - a}{1 - a})^b & \text{if } \theta \in [a, 1] \end{cases} \quad (7)$	Angle to speed mapping. a is a constant between 0 and 1, and b equals 2 or 3. θ is the normalized value ([0, 1]) of the angle of the movement of tracked body parts, which is mapped to the speed of the user's viewpoint in VR (v_v) with k factor.

The Function 1, Function 3 and Function 5 Were Selected for Our Experiment.

[36], [37] (Table 2:2b). The task usually includes an arrow to indicate the direction of the target, and participants need to follow the arrow to reach all targets.

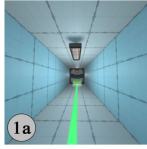
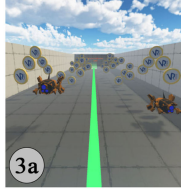
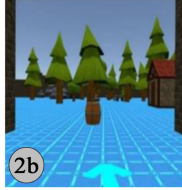
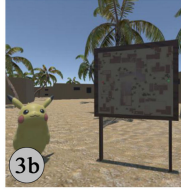
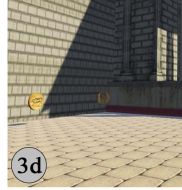
The other task type is locomotion tasks with secondary tasks (Table 2). There are two main secondary tasks used in previous studies: spatial orientation and collecting items. The former task requires participants to point out the direction to the starting position when reaching the target [2], [6], [18], [34], [40]. The latter task asks participants to collect items when navigating in the virtual

environment [1], [10], [18]. These two secondary tasks, in combination of the two locomotion tasks (i.e., follow the path and reach the target), forms four task categories (Table 2:3a-3 d).

3 EXPERIMENT

The goal of this experiment was to investigate the effects of transfer functions and body parts on body-centric locomotion in VR.

TABLE 2
A Summary of Task Design for Body-Centric Locomotion

Movement Path	Task Types		
Straight [1], [4]  1a Curve [1], [18], [20], [46], [51]  1b Line segments [2], [3], [6], [9], [11], [19], [22], [34], [36], [37], [40]  1c	Locomotion tasks without secondary tasks	Locomotion tasks with secondary tasks	
	Follow the path [1], [20], [18], [46], [51]	Follow the path + Collect items [1]	Reach the target + Spatial orientation [2], [6], [34], [40]
	 2a	 3a	 3c
	Reach the target [2], [3], [6], [9], [11], [19], [22], [34], [36], [37], [40]	Reach the target + Collect items [10]	Follow the path + Spatial orientation + Collect items [18]
	 2b	 3b	 3d

(1a) A straight path from [1]. (1b) A curved path from [46]. (1c) A path with line segments from [22]. (2a) Following the path from [20]. (2b) Reaching the target from [22]. (3a) Follow the path + Collect items from [1]. (3b) Reach the target + Collect items from [10]. (3c) Reach the target + Spatial orientation from [34]. (3d) Follow the path + Spatial orientation + Collect items from [18].

3.1 Apparatus

We used a HTC VIVE Pro headset device (refresh rate of 90 Hz and a resolution of 1440×1600 pixels per eye) with two handheld controllers and three HTC trackers for the experiment. The trackers were attached to body parts (i.e., torso and knee) for collecting body movement data. The headset was connected to a computer with an Intel core i7 8700 CPU (3.2 GHz), 16 GB RAM and Geforce GTX-1060 graphics card. The experiment program was developed with Unity 2019 in C#.

3.2 Participants

We recruited 18 participants (9 female) from the local campus, aged from 21 to 25 (M: 22.67, SD: 1.08). Six of them had experience using VR devices. However, none of them had experience with body-centric locomotion in VR. Half of the participants' vision were normal and the other half were corrected to normal. Each of them was paid 8 dollars for the time.

3.3 Experiment Design

The experiment was a within-subject repeated-measures design. The independent variables and dependent variables are as follows.

3.3.1 Independent Variables

The independent variables were body parts for direction control, body parts for speed control and transfer functions.

Body Parts for Direction Control. According to the analysis of related works (Section 2.1), we selected head [17], [37], [40], [55] and torso [20], [28], [31], [38] to control the direction of virtual locomotion (Fig. 2a) in this study.

Body Parts for Speed Control. From our preliminary results [4], the torso was rated as the worst option, mainly due to the tiredness when standing up for torso-based virtual locomotion. We therefore selected head [37], [42], [55], [57], arm [40], [59] and knee [20], [28], [38] to control the speed of virtual movement for this study. Participants leaned forward to perform locomotion with the tracker attached on the chest for data recording [22], and swung their arm for locomotion and movement data were collected via the controller held in their hand [59], and bent their knee for locomotion and the movement data were recorded via the tracker fastened to the knee [51].

For speed control of locomotion with these body parts (Fig. 2a), the displacement (d_r) and the angle (θ) along the yaw axis of body movements (i.e., tilting head, swinging arm, and bending knee) were recorded and standardized, and mapped to the movement speed (v_v) of the user's view-point in VR (Fig. 2b&c). In this experiment, we followed [4] and [18] to set the maximum angle of each body motion as 20 degrees.

Transfer Functions. In this experiment, we chose the specific transfer functions for two main reasons.

First, the selected functions should vary in mapping types. From the literature review, there are three main function types: linear (L), power (P) and piecewise functions

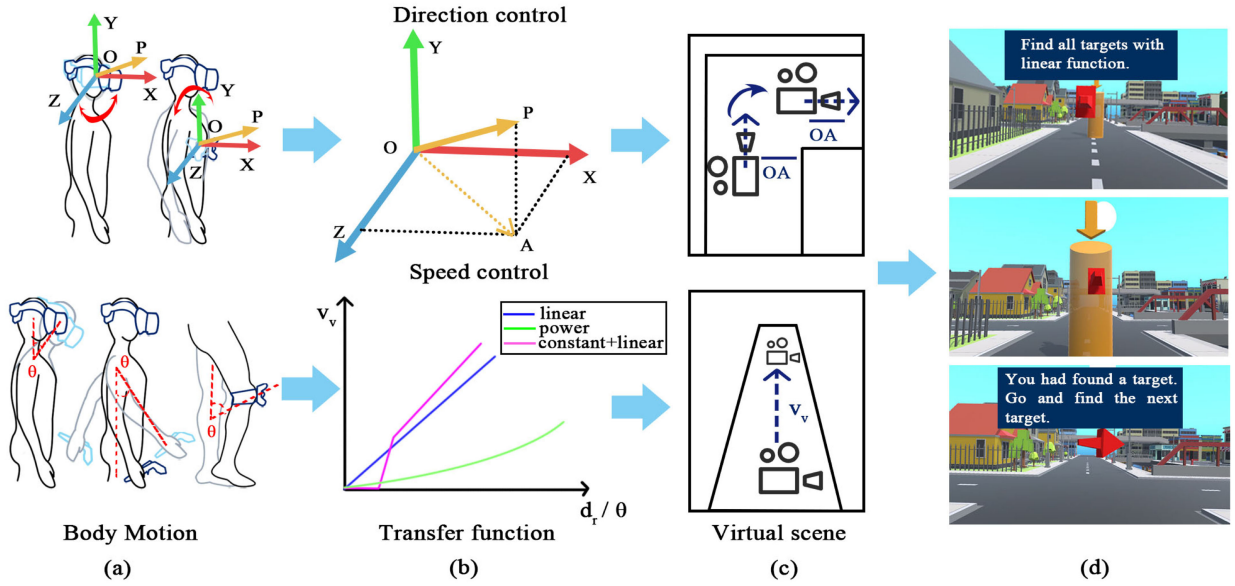


Fig. 2. (a) Top: five body parts for virtual locomotion: head tilt and torso lean for direction control. Bottom: head tilt, arm swing, and knee bend for speed control. (b) Top: the 3 d vector (OP) of headset or tracker is projected onto the x and z plane as the direction of locomotion. Bottom: three types of transfer functions which can convert the angle to the speed of the user's viewpoint. (c) Changes of viewpoint in virtual scene. (d) The user's viewpoint in VR during experimental task. The orange cylinder and arrow indicate a target. The red arrow points out the location for a target, and text hints are present for 2 seconds after starting the trial. When a participant reaches a target, the target disappears after confirmation. The red arrow changes its direction and points to the next target.

with constant and linear/power (CL/CP). Eqs. 1 and 3 belong to the first two function types respectively, and both Eqs. 5 and 7 are in the third type.

Second, our previous work [4], in which we conducted a pilot study to examine the performance of these transfer functions in Table 1, showed that there was no significantly difference on task time between the two linear functions (Eqs. 1 and 2) ($Z = -2.32, p = .26$), between the two power functions (Eqs. 3 and 4) ($Z = -3.15, p = .47$), and between the two piecewise functions (Eqs. 6 and 7) ($Z = -1.46, p = .26$). Hence, we selected one function from each of the above three function pairs: Eqs. 1, 3 and 7. In addition, we included Eq. 5 as this function led to significantly different times than others (all $p < .05$). Thus, four transfer functions (Eqs. 1, 3, 5 and 7) were examined in previous study [4]. Furthermore, the results of our previous study showed that the linear function (L: Eq. (1)) had the highest subjective scores (UEQ-S, UEQ-S pragmatic and UEQ-S hedonic), followed by the power function (P: Eq. (3)) and the piecewise function with constant and linear functions (CL: Eq. (5)). The piecewise function with constant and power functions (CP: Eq. (7)) had the lowest scores and the longest task completion time among the four functions. Overall, we selected the L function, the P function and the CL function for the present study.

The details of the transfer functions are presented in Table 1. Consistent with previous studies [4], [18], [25], [37], k was equal to 3 in Eq. (1), 1.4 in Eq. (3), and 3.5 in Eq. (5) in our experiment. As the angle of body motion for each of transfer functions was standardized (θ ranges from 0 to 1) and mapped to the speed of virtual viewpoint [20], [51], the value of k is the maximum speed of the transfer function. So the maximum speed was 3 m/s for the L function, 1.4 m/s for the P function, and 3.5 m/s for the CL function.

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3.3.2 Dependent Variables

The objective dependent variables were task completion time, traveled distance, environmental collision and the number of overshoot. Task completion time was defined as the time spent reaching all targets on the locomotion path in a trial [18], [20], [22], [34]. Traveled distance referred to the distance that participants traveled in the virtual scene during a trial [6], [11], [18]. Environmental collision was defined as the number of times that participants collided with buildings in the virtual scene in a trial [18], [19]. Number of overshoot was defined as the number of times that the participants entered a target and then left it without selection [11].

The subjective dependent variables were sickness, presence and usability. According to Buttussi *et al.* [22], we used four basic items of SSQ [54] to collect the sense of sickness, including general discomfort (item 1), headache (item 3), eye strain (item 4), and nausea (item 8). We used Slater-Usch-Steed (SUS) questionnaire [43] to measure the sense of presence. In addition, the short version [41] of the user experience questionnaire (UEQ-S) was adopted to measure the pragmatic and hedonic quality of locomotion. *Pragmatic* means that an interaction is simple to understand and learn, easy to interact with, and efficient in completing tasks. *Hedonic* refers to how interesting and exiting an interaction is, and how inventive and leading edge a technology is. Such a questionnaire has been adopted in recent studies of body-centric locomotion [4], [18].

3.3.3 Virtual Scene

We used two virtual scenes in the experiment. One was used for practice purposes (Fig. 3a&c). It was a rectangular area surrounded by four virtual walls [18], [22]. The dimension of the virtual scene was about 20 m $\times$ 30 m. There were three targets along the path with the length of 27 m but no

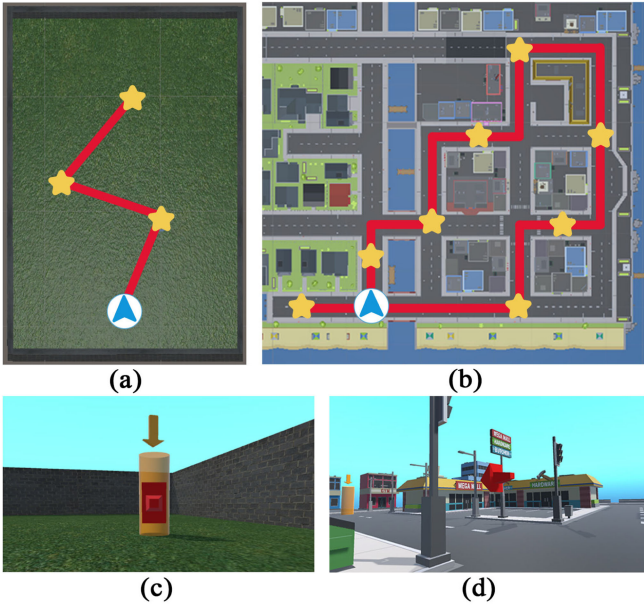


Fig. 3. (a) Top isometric view of the practice scene. The position of the blue arrow is the start point. The stars represent the target locations. Participants need to find all targets. The red path was the route for locomotion. (b) Top isometric view of the experiment scene for the formal test. (c) The virtual viewpoint in the practice scene during practice task. The orange cylinder and arrow indicate a target. (d) The virtual viewpoint in the scene during the formal test. The red arrow indicates the location of a target. Participants need to follow the red arrow to reach the target.

obstacles in the scene (Fig. 3a). The other scene was used in the formal experiment (Fig. 3b&d). It was a virtual city model with streets and buildings [3], [9], [20], [51] (Fig. 3d). The dimension of the formal scene was about 120 m $\times$ 108 m. The length of the streets was 260 m and the width of streets was about 5 m [18].

An orange cylinder with a radius of 0.5 m [4], [6] was used to represent a target (Fig. 3c). An orange arrow above the cylinder indicated the target location [4], [19], [40]. The color of the target changed depending on the distance from the participant to the target [19]. When the participant was outside the target, its color was orange. Otherwise, it changed to green.

A red arrow floated at eye level indicated the direction of the target [19], [34] (Fig. 3c&d). The arrow could provide hints about the target's location. When the participant reached a target, the red arrow would shift towards the next target.

3.4 Procedure

At the beginning of the experiment, participants were asked to fill in a pre-test questionnaire about sickness and sign a consent form. The experiment consisted of 6 combinations of 2 body parts for direction control and 3 body parts for speed control. The order of 6 combinations was counter-balanced by Latin square. For each combination, the experimental procedure included 3 phases (practice phase, formal test, and subjective feedback collection).

In the practice phase, participants were informed about the procedure of the practice task and put on the VR headset. The main purpose of practice was to let the participants be familiar with the locomotion tasks with specified body parts. The practice task required participants to begin from the starting point and reach each of three targets indicated

by the arrow. The participants needed to press the trigger of the controller when reaching a target.

In the formal test phase, participants were asked to perform the formal locomotion task while standing. During the formal test, the locomotion methods were the same with the practice phases, except with the different virtual scenes. For each body combination, participants to complete tasks with the 3 transfer functions. For each transfer function, participants needed to reach all 8 targets (Fig. 3). Participants started a task by pressing the trigger of controller, and followed the direction indicated by the red arrow to find the targets one by one. When participants reached a target, the color of the target changed from orange to green, prompting them to press the trigger of the controller for confirmation. Then the arrow would point to the next target. Participants were asked to complete the task as accurately and quickly as possible.

After completing the task for a transfer function with a specified body combination, participants were asked to fill in a questionnaire. We did not ask participants to complete the questionnaire after finishing the entire experiment, as participants may get confused about the six body combinations and the three transfer functions. Participants can adjust their subjective ratings during the entire experiment.

In summary, the experiment consisted of (excluding practice trials) 18 participants $\times$ 2 direction control $\times$ 3 speed control $\times$ 3 transfer functions = 324 trials. The entire experiment lasted about two hours for each participant (the practice and formal tasks took about 1 h, and the remaining time was for the rest and subjective evaluation).

4 RESULTS AND ANALYSIS

We collected a total of 324 trials $\times$ 8 targets = 2592 trials data. We removed 29 outliers (about 1.12% of the data) with more than three standard deviations from the mean time for further analysis. The *Shapiro-Wilk* method was used to test the normality of the experimental data. When experimental data satisfied the conditions of normality, the data were analyzed with repeated-measures ANOVA with *Bonferroni correction* applied to post-hoc test. Whenever Mauchly's test indicated that the assumption of sphericity had been violated ($p < 0.05$), the degrees of freedom were corrected using Greenhouse-Geisser or Huynh-Feldt estimates. Otherwise, data analysis was conducted with *Generalized Estimating Equations* and a *Bonferroni correction* for post-hoc test.

4.1 Task Completion Time

Body parts for direction control had a significant main effect on task completion time ($F_{1,17} = 4.90$, $p < .05$, $\eta_p^2 = .22$) (Fig. 4) Table 3. Post-hoc test showed that the head-based direction control (M: 150.69 s, SD: 59.08 s) had a shorter completion time than the torso-based direction control (M: 155.63 s, SD: 55.70 s) ($p < .05$). However, body parts for speed control had no significant main effect on task completion time ($F_{2,34} = 2.92$, $p = .07$, $\eta_p^2 = .15$) (Fig. 4). The mean value was 156.60 s (SD = 60.33 s) for head, 152.23 s (SD = 62.96 s) for arm, and 150.65 s (SD = 48.13 s) for knee. Transfer functions had a significant main effect on task completion time ($F_{2,34} = 2215.18$, $p < .01$, $\eta_p^2 = .99$) (Fig. 4). Post-hoc test showed that the CL function had the shortest time (M: 107.59 s, SD: 19.74 s), followed by the L function (M:

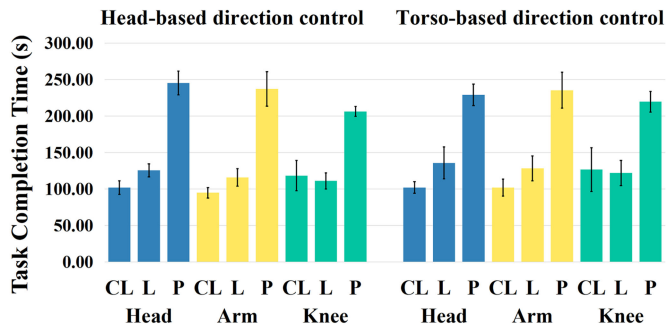


Fig. 4. Task completion time for two body parts for direction control, three body parts for speed control and three types of transfer functions. (L: linear function; P: power function; CL: piecewise function with constant and linear functions). Error bars represent SD. The abbreviations of the three functions are also applied to the rest figures and tables.

123.04 s, SD: 17.03 s), and the P function had the longest time (M: 228.85 s, SD: 21.59 s) (all $p < .01$).

There were interaction effects on task completion time among body parts for direction control and speed control and transfer functions ($F_{2,840,48.273} = 2.94, p < .05, \eta_p^2 = .15$).

In the head-based direction control, there was an interaction effect between transfer functions and body parts for speed control ($F_{2,463,41.869} = 31.81, p < .01, \eta_p^2 = .65$). We further performed a simple main effect analysis for body parts for speed control. For each of the three body parts, there was a significant main effect for transfer functions on task completion time (head: $F_{1,413,24.327} = 1360.31, p < .01, \eta_p^2 = .99$; arm: $F_{1,491,25.342} = 499.18, p < .01, \eta_p^2 = .97$; knee: $F_{1,449,24.631} = 281.27, p < .01, \eta_p^2 = .94$) (Fig. 4). Post-hoc tests revealed that task time significantly increased from CL, L to P functions (all $p < .01$) when participants used head or arm to control speed. However, when they used a knee to

control speed, there was no significant difference between CL and L functions ($p = .53$), but both of them had shorter times than that of P function (both $p < .01$).

We also conducted the analysis of simple main effects for each transfer function. There was a significant main effect for speed control using body parts on task completion time (Fig. 4) (CL: $F_{2,34} = 18.47, p < .01, \eta_p^2 = .52$; L: $F_{2,34} = 12.49, p < .01, \eta_p^2 = .42$; P: $F_{2,34} = 37.17, p < .01, \eta_p^2 = .69$). Post-hoc tests showed that task time with head or arm was shorter than that with a knee (both $p < .01$) for CL function, but no significant difference between head and arm was found ($p = .07$). In contrary with CL function, task time with knee was shorter than that with the head or arm for P function (both $p < .01$), and there was also no significant difference between head and arm ($p = .33$). For L function, task time with arm ($p < .05$) or knee ($p < .01$) was shorter than that with head. And no significant difference between arm and knee was found ($p = .59$).

In the torso-based direction control, there was an interaction effect between body parts for speed control and transfer functions ($F_{2,447,41.601} = 11.76, p < .01, \eta_p^2 = .41$). We further performed a simple main effect analysis for each of three body parts with speed control (Fig. 4). We found that the trend of the main effect was the same as that in head-based direction control. Transfer functions had a significant main effect on task completion time for each of the body parts (head: $F_{2,34} = 505.66, p < .01, \eta_p^2 = .97$; arm: $F_{1,521,25.850} = 470.89, p < .01, \eta_p^2 = .97$; knee: $F_{1,467,24.940} = 149.29, p < .01, \eta_p^2 = .90$). Task time significantly increased from CL, L to P functions (all $p < .01$) when participants used head or arm to control speed. However, when they used knee to control speed, the task time for both CL and L functions was shorter than that of P function (both $p < .01$), and no significant difference between CL and L functions was found ($p = 1.00$).

TABLE 3

Mean Values (M) and SD of the Objective and Subjective Performance Measures for Two Body Parts for Direction Control, Three Body Parts for Speed Control and Three Types of Transfer Functions

Conditions		Head-based direction control									Torso-based direction control								
		Head			Arm			Knee			Head			Arm			Knee		
		L	P	CL	L	P	CL	L	P	CL	L	P	CL	L	P	CL	L	P	CL
Task Completion Time (s)	M	125.53	245.44	101.87	115.70	237.17	94.76	111.08	206.33	118.31	135.79	229.03	101.96	128.30	235.45	101.98	121.86	219.65	126.64
	SD	8.94	16.28	9.21	12.02	23.52	7.24	11.10	6.83	20.79	21.85	14.66	7.89	16.95	24.78	11.63	17.21	14.19	30.05
Traveled Distance (m)	M	297.51	305.45	298.54	275.43	278.69	284.10	285.27	273.32	300.84	290.66	277.55	294.40	281.74	277.74	300.17	297.06	283.93	317.86
	SD	14.30	17.87	22.52	13.51	15.76	14.59	17.50	7.16	34.74	25.98	10.24	15.86	15.27	12.07	24.62	21.60	14.33	36.18
Overshoot	M	0.03	0.01	0.12	0.01	0.01	0.11	0.07	0.00	0.28	0.08	0.01	0.12	0.07	0.01	0.18	0.09	0.02	0.11
	SD	0.07	0.04	0.16	0.06	0.03	0.13	0.13	0.00	0.35	0.16	0.06	0.15	0.18	0.03	0.18	0.12	0.06	0.13
Environmental collisions	M	0.33	0.22	1.00	0.17	0.00	0.67	0.39	0.33	3.17	3.28	0.28	2.94	1.00	0.06	5.11	7.28	0.89	9.94
	SD	0.84	0.65	2.00	0.51	0.00	1.94	1.20	1.03	5.52	6.41	0.96	6.02	3.29	0.24	10.48	14.07	1.88	12.36
General Discomfort	M	0.06	0.17	0.33	0.11	0.11	0.44	0.11	0.06	0.22	0.17	0.22	0.28	0.11	0.00	0.33	0.28	0.11	0.39
	SD	0.24	0.38	0.77	0.32	0.32	0.86	0.47	0.24	0.55	0.51	0.55	0.57	0.47	0.00	0.97	0.57	0.32	0.85
Headache	M	0.06	0.06	0.17	0.06	0.06	0.33	0.06	0.00	0.11	0.11	0.22	0.17	0.11	0.00	0.11	0.22	0.11	0.33
	SD	0.24	0.24	0.38	0.24	0.24	0.59	0.24	0.00	0.32	0.47	0.65	0.51	0.47	0.00	0.32	0.55	0.32	0.77
Eyestrain	M	0.22	0.44	0.33	0.17	0.28	0.17	0.17	0.28	0.17	0.28	0.28	0.33	0.22	0.11	0.17	0.33	0.28	0.39
	SD	0.43	0.62	0.59	0.38	0.46	0.38	0.38	0.57	0.38	0.57	0.57	0.59	0.55	0.32	0.38	0.49	0.46	0.78
Nausea	M	0.11	0.22	0.44	0.06	0.11	0.44	0.06	0.06	0.17	0.11	0.11	0.28	0.22	0.06	0.44	0.28	0.17	0.39
	SD	0.32	0.55	0.92	0.24	0.32	0.70	0.24	0.24	0.51	0.47	0.47	0.57	0.55	0.24	0.78	0.75	0.51	0.78
Presence	M	2.44	2.56	2.50	2.39	2.61	2.00	1.89	1.83	1.78	2.11	2.22	1.67	2.61	2.61	2.00	1.94	1.94	1.83
	SD	2.83	2.79	2.64	2.55	2.73	2.54	2.27	2.28	2.29	2.76	2.80	2.52	2.77	2.77	2.59	2.39	2.41	2.38
UEQ-S	M	4.79	4.16	4.83	5.24	4.40	4.10	4.72	4.00	4.67	4.75	3.99	4.65	5.52	4.52	4.72	4.89	4.27	4.67
	SD	1.66	1.74	1.61	1.05	1.65	1.74	1.37	1.54	1.34	1.42	1.65	1.58	0.90	1.41	1.35	1.38	1.45	1.45
UEQ-S pragmatic	M	4.92	4.26	5.06	5.26	4.57	3.96	4.68	3.97	4.46	4.64	4.03	4.50	5.65	4.61	4.49	4.68	4.00	4.35
	SD	1.77	1.70	1.71	1.06	1.54	1.68	1.31	1.52	1.51	1.36	1.51	1.43	1.07	1.30	1.58	1.62	1.39	1.72
UEQ-S hedonic	M	4.67	4.06	4.60	5.22	4.22	4.25	4.76	4.03	4.88	4.86	3.96	4.79	5.39	4.43	4.96	5.10	4.54	5.00
	SD	1.74	1.91	1.67	1.22	1.83	1.93	1.59	1.78	1.46	1.61	1.93	1.82	0.97	1.71	1.37	1.39	1.70	1.40

Gray color indicates the maximum and minimum values.

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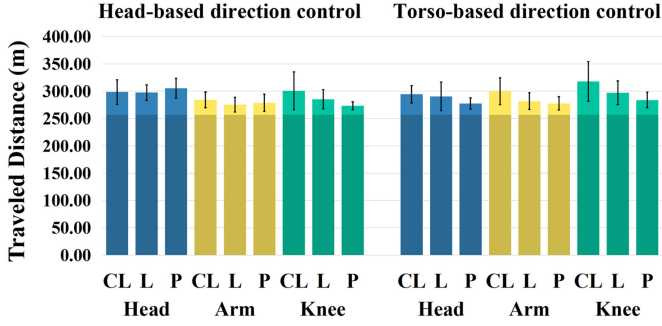


Fig. 5. Traveled Distance for two body parts to control direction, three body parts to control speed and three types of transfer functions. The dashed areas of the bars in the histogram mean the shortest distance (260 m).

The analysis of a simple main effect for each transfer function was also conducted. There was a significant main effect for both CL and P functions on task completion time (CL: $F_{1,182,20.098} = 9.33, p < .01, \eta_p^2 = .35$; P: $F_{2,34} = 5.28, p < .05, \eta_p^2 = .24$). Post-hoc tests showed that task time with head or arm was shorter than that with a knee (both $p < .05$) for CL function, but no significant difference between head and arm was found ($p = 1.00$). In contrast, task time with knee was shorter than that with arm for P function ($p < .01$). However, no significant difference was found between head and arm ($p = .79$), and between head and knee ($p = .16$). For L function, no significant main effect was found (L: $F_{2,34} = 2.91, p = .07, \eta_p^2 = .15$).

4.2 Traveled Distance

There was no significant main effect for the body parts for direction control on traveled distance ($F_{1,17} = 1.33, p = .27, \eta_p^2 = .07$) (Fig. 5). The mean traveled distance were 288.79 m (SD: 21.69 m) for head, and 291.24 m (SD: 23.93 m) for torso.

However, the body parts for speed control had a significant main effect on traveled distance ($F_{2,34} = 7.85, p < .01, \eta_p^2 = .32$). Participants could travel shorter distance with arm (M: 282.98 m, SD: 18.06 m) than that with head ($p < .01$) (M: 294.02 m, SD: 20.07 m) or knee ($p < .05$) (M: 293.05 m, SD: 27.74 m) in VR, and no significant effect between head and knee was found ($p = 1.00$).

As with the body parts for speed control, transfer functions had a significant main effect on traveled distance ($F_{1,378,23.428} = 23.14, p < .01, \eta_p^2 = .58$). The P function had the shortest distance traveled (M: 282.78 m, SD: 16.86 m), followed by the L function (M: 287.94 m, SD: 19.83 m), and the CL function had the longest distance traveled (M: 299.32 m, SD: 27.42 m) (all $p < .05$).

No interaction effect among the body parts for direction control and speed control and transfer functions on traveled distance ($F_{4,68} = 1.20, p = .32, \eta_p^2 = .07$) was found. However, there was an interaction effect between the body parts for speed control and transfer functions on traveled distance ($F_{2,198,37.373} = 4.95, p < .05, \eta_p^2 = .23$).

We therefore further performed the analysis of simple main effects for each body part for speed control. Arm ($F_{2,34} = 15.47, p < .01, \eta_p^2 = .48$) or knee ($F_{2,34} = 14.17, p < .01, \eta_p^2 = .46$) had significant main effects for transfer functions on traveled distance, while head had no such effect ($F_{2,34} = 0.86, p = .43, \eta_p^2 = .05$). Post-hoc tests showed that traveled

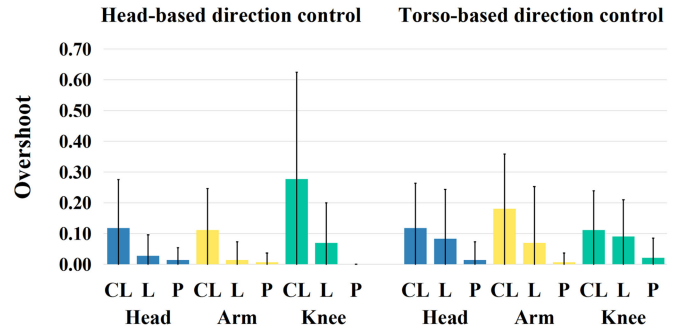


Fig. 6. Overshoot for two body parts for direction control, three body parts for speed control and three types of transfer functions.

distance with L (M: 278.58 m, SD: 14.56 m) or P function (M: 278.22 m, SD: 13.84 m) was smaller than that with CL (M: 292.13 m, SD: 21.55 m) (both $p < .01$) for arm, but no significant difference between L and P function was found ($p = 1.00$). However, traveled distance with P (M: 278.62 m, SD: 12.39 m) was smaller than that with L (M: 291.17 m, SD: 20.28 m) or CL (M: 309.35 m, SD: 36.01 m) function for knee (both $p < .01$), and no significant difference between L and CL function existed ($p = .07$).

We also conducted an analysis of simple main effects for each transfer function. P ($F_{2,34} = 13.94, p < .01, \eta_p^2 = .45$) and L ($F_{2,34} = 9.97, p < .01, \eta_p^2 = .37$) had a significant main effect for body parts for speed control on traveled distance. However, CL had no significant main effect for body parts for speed control (CL: $F_{1,444,24.551} = 3.61, p = 0.06, \eta_p^2 = .18$). Post-hoc tests showed that traveled distance with arm (M: 278.22 m, SD: 13.84 m) or knee (M: 278.62 m, SD: 12.39 m) was smaller than that with head (M: 291.50 m, SD: 20.15 m) (both $p < .01$) for P function, but no significant difference between arm and knee was found ($p = 1.00$). For L function, traveled distance with arm (M: 278.58 m, SD: 14.56 m) was smaller than that with head (M: 294.08 m, SD: 20.96 m) or knee (M: 291.17 m, SD: 20.28 m) (both $p < .01$), and there was no significant difference between head and knee ($p = 1.00$).

4.3 Overshoot

Due to the violation of normality test on overshoot across participants, we used *Generalized Estimating Equations* test with a *Bonferroni correction* to analyze the difference of body parts for direction control and speed control, and transfer functions in terms of overshoot.

Body parts (head (M: .07, SD: .17) versus torso (M: .08, SD: .14)) for direction control had no significant main effect for overshoot ($\chi^2(1) = .12, p = .73$) (Fig. 6).

There was also no significant main effect for body parts for speed control head (M: .06, SD: .12), arm (M: .06, SD: .14) and knee (M: .10, SD: .19) on overshoot ($\chi^2(2) = 4.75, p = .09$).

However, transfer functions had a significant main effect on overshoot ($\chi^2(2) = 61.58, p < .01$) (Fig. 6). Post-hoc test showed that the P function had the smallest number of times for overshoot (M: .01, SD: .04), followed by the L function (M: .06, SD: .13), and the CL function had the largest number of times (M: .15, SD: .20) (all $p < .01$).

There was an interaction effect among body parts for direction control and speed control and transfer functions on overshoot ($\chi^2(4) = 10.22, p < .05$).

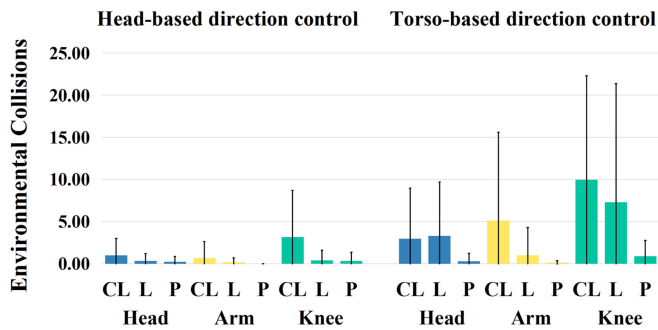


Fig. 7. Environmental collisions for two body parts for direction control, three body parts for speed control and three types of transfer functions.

In the head-based direction control, an interaction effect between body parts for speed control and transfer functions was found ($\chi^2(4) = 18.86, p < .01$). We further performed a simple main effect analysis for speed control using head ($\chi^2(2) = 9.53, p < .01$), arm ($\chi^2(2) = 16.21, p < .01$), and knee ($\chi^2(2) = 20.00, p < .01$). Post-hoc tests revealed that the number of times for overshoot was significantly larger with CL function, compared with both L ($p < .05$) and P ($p < .01$) functions when participants used head or arm to control speed. However, no significant difference existed between L and P functions for head ($p = 1.00$) and arm ($p = .91$). CL function had a larger the number of times for overshoot than P function when participants used knee to control speed ($p < .01$). However, no significant difference existed between L and P functions, CL and L functions (both $p = 0.06$) for knee.

We also conducted the analysis of a simple main effect for each transfer function. L function had a significant main effect for speed control using body parts on overshoot (Fig. 6) (L: $\chi^2(2) = 7.31, p < .05$). However, Post-hoc tests showed that no significant difference for each pair of body parts (head versus arm ($p = .40$), head versus knee ($p = .54$), arm versus knee ($p = .15$)) on overshoot. For both CL and P functions, there was no significant main effect for body parts for speed control on overshoot (CL: $\chi^2(2) = 5.74, p = .06$, $\chi^2(2) = 3.60, p = .17$).

In the torso-based direction control, no interaction effect between speed control using body parts and transfer functions was found ($\chi^2(4) = 4.39, p = .36$) (Fig. 6).

4.4 Environmental Collisions

As with overshoot, we used *Generalized Estimating Equations* test with a *Bonferroni correction* for the analysis of environmental collision.

Body parts for direction control had a significant main effect on the number of collision ($\chi^2(1) = 15.01, p < .01$) (Fig. 7). The average number of collision with head (M: .70, SD: 2.30) was significantly smaller than that with torso (M: 3.42, SD: 8.30). Most probably, the participants found that the head-based direction control was well coupled with their viewpoint in VR, which led to smaller number of collisions.

Body parts for speed control had a significant main effect on the number of collision ($\chi^2(2) = 7.08, p < .05$) (Fig. 7). Post-hoc test showed that the average number of environmental collisions with head (M: 1.34, SD: 3.86) was smaller than that with knee (M: 3.67, SD: 8.68) ($p < .05$), while no significant effect was found between head and arm ($p = 1.00$), and between arm and knee ($p = .08$).

Transfer functions also had a significant main effect on the number of environmental collisions ($\chi^2(2) = 32.01, p < .01$) (Fig. 7). Post-hoc test showed the average number of environmental collisions with both L (M: 2.07, SD: 6.83) and P (M: .30, SD: 1.02) functions was smaller than that with CL (M: 3.81, SD: 7.96) function (all $p < .05$). However, no significant difference existed between L and P functions ($p = .07$).

No interaction effect on the number of environmental collisions among body parts for direction control and speed control and transfer functions ($\chi^2(4) = 5.52, p = .24$) was found. In addition, there was no significant interaction effect between speed control using body parts and transfer functions on the number of environmental collisions ($\chi^2(4) = 8.02, p = .09$).

4.5 Sickness

We followed the way of [22] to analyze the effects of body parts for direction control, body parts for speed control and transfer functions on general discomfort, headache, eye-strain and nausea. There were no significant main effects for body parts for direction control, body parts for speed control and transfer functions on general discomfort and eye-strain (Table 4).

In addition, only transfer function had significant main effects on both headache and nausea. Post-hoc test showed participants felt more headache with CL function than P ($p < .01$), but there was no significant difference between L and CL ($p = .10$), between L and P ($p = .72$). As with headache, participants felt more nausea when using CL, compared with L ($p < .05$) and P ($p < .01$), but no significant difference was found between L and P ($p = .91$).

4.6 Presence

As with previous work [22], there were no significant main effects for body parts for direction control, body parts for speed control, and transfer functions on presence (Table 4), according to the analysis of SUS questionnaire [43].

4.7 Usability

We analyzed the effects of direction control and speed control using body parts and transfer functions on UEQ-S, UEQ-S *pragmatic* and UEQ-S *hedonic* [4], [18], [41] (Table 4 and 3).

UEQ-S. Both body parts for direction control and body parts for speed control had no significant main effect on UEQ-S. However, transfer functions had a significant main effect on UEQ-S. Post-hoc test showed that participants rated higher UEQ-S score for L function than that for CL or P functions (both $p < .01$), thanks to the suitable speed for L function from participants' feedback, but no significant difference between CL and P function ($p = .09$).

UEQ-S Hedonic. Unlike UEQ-S, body parts for direction control had significant main effects on UEQ-S *Hedonic*. Post-hoc test showed that the participants rated higher UEQ-S *Hedonic* value with torso than that with head ($p < .05$). However, body parts for speed control did not have a significant main effect on UEQ-S.

Transfer functions had a significant main effect on UEQ-S *Hedonic*. Post-hoc test showed that L and CL had a higher *Hedonic* than P (both $p < .05$), but no significant difference between L and CL ($p = .078$).

UEQ-S Pragmatic. As with UEQ-S, both body parts for direction control and body parts for speed control had no

TABLE 4
Statistical Significance of Subjective Items (D: Body Parts for Direction Control, S: Body Parts for Speed Control, F: Transfer Function)

Item	Main Effect			Interaction Effect	
		D	S	F	D * S * F
General Discomfort	χ^2	$\chi^2(1) = .17$	$\chi^2(2) = .06$	$\chi^2(2) = 5.85$	$\chi^2(4) = 1.20$
	p	0.68	0.97	0.05	0.88
Headache	χ^2	$\chi^2(1) = .96$	$\chi^2(2) = .16$	$\chi^2(2) = 11.88$	$\chi^2(4) = 6.37$
	p	0.33	0.92	< .01	0.17
Eyestrain	χ^2	$\chi^2(1) = .23$	$\chi^2(2) = 4.88$	$\chi^2(2) = 2.07$	$\chi^2(4) = .29$
	p	0.63	0.09	0.36	0.99
Nausea	χ^2	$\chi^2(1) = .33$	$\chi^2(2) = .13$	$\chi^2(2) = 8.86$	$\chi^2(4) = 1.86$
	p	0.57	0.94	< .05	0.76
Presence	χ^2	$\chi^2(1) = .38$	$\chi^2(2) = 2.86$	$\chi^2(2) = 2.11$	$\chi^2(4) = 3.77$
	p	0.54	0.24	0.35	0.44
UEQ-S	χ^2	$\chi^2(1) = .84$	$\chi^2(2) = 1.90$	$\chi^2(2) = 19.67$	$\chi^2(4) = 2.64$
	p	0.36	0.39	< .01	0.62
UEQ-S hedonic	χ^2	$\chi^2(1) = 3.92$	$\chi^2(2) = .99$	$\chi^2(2) = 12.67$	$\chi^2(4) = 4.06$
	p	< .05	0.61	< .01	0.40
UEQ-S pragmatic	χ^2	$\chi^2(1) = .02$	$\chi^2(2) = 4.04$	$\chi^2(2) = 22.32$	$\chi^2(4) = 2.42$
	p	0.90	0.13	< .01	0.66

significant main effect on UEQ-S *Pragmatic*. However, a significant main effect for transfer functions on UEQ-S *Pragmatic* was found. Post-hoc test showed that L had a higher *Pragmatic* value than CL and P (both $p < .01$), but no significant difference between CL and P ($p = .70$).

In addition, there was no significant interaction effect among body parts for direction control and body parts for speed control and transfer functions on three measures. However, speed control using body parts had a significant interaction effect with transfer functions on three measures (Table 4).

5 DISCUSSION

5.1 Body Parts for Direction Control

We analyzed the effects of two body parts (i.e., head and torso) for direction control on virtual locomotion in terms of task completion time, traveled distance, overshoot, environmental collision and subjective rating. In general, head-based direction control was more efficient and easier to manipulate than the torso, according to the analysis of task completion time and the number of environmental collisions. The participants reported that the way of controlling direction with head was well coupled with eye viewing, so may enable a fast and precise locomotion performance. This result was consistent with Williams *et al.* [7], which showed that participants' spatial orientation was better when translated forward with the direction they were looking in VR.

However, the use of head negatively impacted the experience of enjoyment when rotating the head to view the virtual environment, according to the analysis of the UEQ-S *Hedonic*. Consistent with [7], [8], the participants could only look in the forward direction when performing virtual locomotion with head, but they could look around when using torso to control direction. Therefore, the use of the head to control direction may not be suitable for looking around while moving in VR. For such a case, participants can stop movement to look around in VR, as suggested in the study of Zielasko *et al.* [18].

The accuracy for direction control did not vary between the two body parts in traveled distance and the number of

overshoot. This result was contrary to the conclusion of the work [18], which showed that torso-based direction control performed the best. Most probably, the contributing factor which led to such different results was the experimental setting (standing versus sitting). In our study, participants may feel tired when performing locomotion using torso while standing. However, sitting on the chair helped participants coordinate the relative position of torso and head when turning, which reduced the difficulty of torso-based direction control.

In summary, the use of head to control direction was more efficient and led to smaller number of environmental collisions than the torso. Even though the torso-based direction control was rated with a higher UEQ-S *Hedonic*, the participants may pause their movement to look around when using head to control direction in virtual environments [18]. Therefore, the use of head to control direction when performing virtual locomotion while standing is recommended.

5.2 Body Parts for Speed Control

In line with previous work [4], [51], the three body parts for speed control did not vary in task completion time for virtual locomotion, indicating that participants could equally utilize these body parts to control speed of virtual locomotion.

In this study, we adopted more objective (i.e., overshoot, traveled distance, environmental collision) and subjective measures (i.e., motion sickness, presence) to thoroughly evaluate the locomotion performance. The performance of body-centric locomotion varied among the three body parts for speed control, according to the analysis of traveled distance and the number of environmental collisions.

The traveled distance with arm-based speed control was shorter than that with head or knee, indicating that the participants could easily use the arm to control locomotion speed. As the arm-based speed control was as natural as daily walking [40], [59], the participants only needed to swing their arms to adjust locomotion speed. For the head-based speed control, participants reported that the long-time usage of a heavy headset with head tilting would cause discomfort and lead to longer traveled distance. The use of knee for speed control was not natural enough. In particular, it was not easy to precisely manipulate how many degrees the knee bent for viewpoint updates, resulting in longer traveled distance as well.

For the environmental collision, the knee-based speed control led to a larger number of environmental collisions than head-based. Such a behavior was hard to perform for a long time while standing, which made the participants feel tiredness within a longer experiment duration. However, existing work [17], which adopted this type of knee-based speed control in a sitting pose, could achieve better performance for virtual locomotion. The sitting pose may contribute to this knee-based speed control for locomotion.

According to the above analysis, the arm is suitable for manipulating the speed of locomotion while standing, which was reinforced the benefits of the arm-swinging method [15], [40], [59]. However, such a result was inconsistent with our previous study [4], which recommended the head as the highest priority. This may be due to the difference of locomotion tasks between the two studies. Our previous study only used body parts for speed control with

straight paths [4]. This study adopted body parts for both speed and direction control in locomotion with line segments, so the participants may turn direction with head while manipulating speed using arm.

5.3 Effects of Transfer Functions on Locomotion

As our previous work [4], the present study also showed that transfer functions played an important role in body-centric locomotion. It was found that transfer functions had significant main effects on measures such as task completion time, traveled distance, the number of overshoots, the number of environmental collisions, the level of motion sickness and UEQ-S. In general, task completion time significantly decreased from the P, L to CL functions, while traveled distance and the number of overshoots increased from the P, L to CL functions.

Virtual locomotion with the CL function was fast but error-prone. There are two main reasons for this. First, the maximum locomotion speed with the CL function is larger than the average speed of common gait-free body-centric locomotion techniques [45], [48]. Comfortable and suitable locomotion speeds are usually set between 1.4 m/s and 3 m/s for gait-free body-centric locomotion techniques in VR (the average speed is 1.78-2 m/s) [9], [44], [45]. However, the maximum locomotion speed with the CL function in this study was 3.5 m/s [4], [37], which was predefined as the value of k in the CL function due to the standardized of inputting from body motion (i.e., angle or displacement). The CL function caused more nausea and headache than the P and L functions due to higher locomotion speed. Second, the subjective evaluation can partly reflect the performance of using the CL function with body parts for locomotion performance. For example, for the combination of the CL function and knee for speed control, most participants reported that the locomotion speed was fast, but it was hard to control precisely. In particular, they could not stop in time when reaching the target, leading to a larger number of environmental collisions and overshoots than that with arm or head. In addition, the CL function with the arm or head for speed control was acceptable to some participants. They mentioned that *"I preferred the CL function with arm, as it gives me the feeling of 'flying' and helps me to reach the target quickly"*.

Virtual locomotion with the P function was easy to control but slow. The main reason lies in the locomotion speed with the P function in VR. The maximum speed with the P function (1.4 m/s) [4] was smaller than the average of visual walking speed in VR [9], [45]. The participants then could easily control the locomotion speed for accurate performance with the cost of longer completion times. While the locomotion with the P function could lead to a smaller number of overshoots and environmental collisions and shorter traveled distance, it made the participants feel boring during locomotion.

Unlike the two functions above, virtual locomotion with the L function was suggested as the tradeoff between accuracy and efficiency for two main reasons. First, the average locomotion speed with the L function was close to the average speed of common gait-free body-centric locomotion techniques (1.78-2 m/s) [9], [44], [45]. The subjective experience of using the L function was comfortable and operable, so the L function was rated higher than other functions in UEQ-S pragmatic and UEQ-S. The participants could easily perform tasks with

the L function and the head or arm to achieve comfortable and operable locomotion. Second, many combinations of body parts for direction and speed control were well coupled with the L function. For example, using head-based direction control, arm-based speed control with the L function could provide the best locomotion performance among all combinations. With torso-based direction control, arm-based speed control with the L function could also achieve acceptable performance in terms of task time and accuracy.

5.4 Effects of Task Types on Locomotion

While our literature review summarizes two main task types (locomotion tasks with or without secondary tasks), our experiment design only adopted the type of locomotion without secondary tasks (i.e., reaching targets in segment paths) for two reasons. First, such a design is representative in VR locomotion research, such as [2], [3], [6], [9], [11], [19], [22], [34], [36], [37], [40]. Second, the aim of our study was to investigate the effects of body parts and transfer functions on virtual locomotion. Involving secondary tasks in experiment design needs to consider additional factors such as secondary task types and the way of performing secondary tasks, which may complicate the experiment and possibly confuse the results.

Admittedly, using one task type leads to a major limitation of our work, that is, unable to look into the effects of task types on body-centric locomotion performance. Instead, we discuss the results of the present paper in comparison to our VR conference paper [4], which investigated virtual locomotion with the task of following straight paths. Apart from the locomotion task type, the two studies are similar in experimental design, hence would enable a relatively fair comparison to discuss the effects of task types on body-centric locomotion.

On one hand, there was some consistency in locomotion performance for the two tasks in our two studies. First, for each task, the body parts for speed control did not vary in task completion time. The body parts used between two tasks were consistent, and the participants could adopt each of the three body parts to complete the locomotion task in similar time. Second, the L function was recommended with the highest priority for both tasks, because of the trade-off between accuracy and efficiency, as well as between comfortable and easy to control.

On the other hand, the combinational effects on locomotion performance were inconsistent between the two tasks. For the task of following straight paths, the head for speed control with the L function was considered as the highest priority for suitable locomotion. Head tilting is tightly coupled with user's viewpoint in VR, so could support a fast and precise locomotion performance in straight path. However, torso-based locomotion should be avoided due to the tiredness of long-time standing during the experiment. For the task of reaching targets in segment paths, the combination of head for direction control, arm for speed control and the L function was recommended as the highest priority. As the head is tightly coupled with users' viewpoint, they could easily change direction by rotating the head. In addition, using the arm for speed control is straightforward, as it is natural to swing arms with two controllers held in hand. So the combination of the two body parts led to the best locomotion performance.

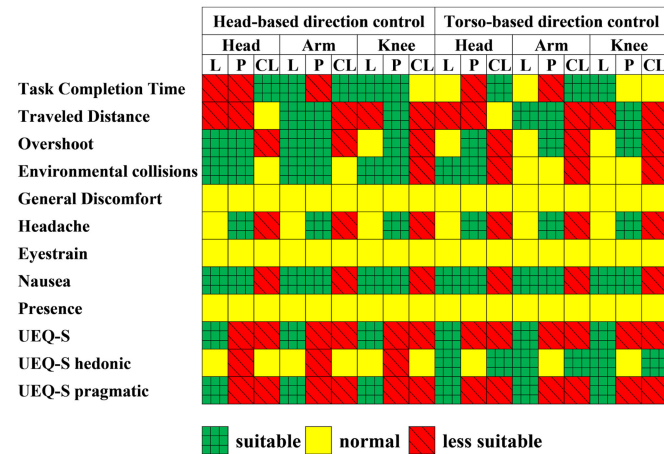


Fig. 8. Visualization of quantitative evaluation for two body parts for direction control, three body parts for speed control and three types of transfer functions.

5.5 Combinational Effects on Locomotion Performance

The different combinations of body parts and transfer functions may lead to different locomotion performances in VR. In summary, the combination of body parts for direction and speed control with transfer functions had significant interaction effects on both task completion time and overshoot. In addition, body parts for speed control and transfer functions had interaction effects on task completion time, overshoot, traveled distance, headache, nausea, and usability.

In order to explicitly analyse the combinational effects, we followed the judgement of “suitable,” “normal” and “less suitable” proposed by [4], [51]. The suitability is judged based on the statistical analysis, such that a combination of body parts for direction control and speed control with transfer function which are significantly better than others are treated as “suitable,” those which are significantly worse than others are judged as “less suitable,” and the others are judged as “normal.” As a summary, the quantitative evaluation results as a colored table are illustrated in Fig. 8. This colored table with the judgements was consistent with quantitative evaluation.

In order to describe the evaluation clearly, we defined a combination of body parts with transfer functions as [D: the way of direction control, S: the way of speed control, F: transfer function]. Based on these judgement rules, we regarded the combination of [D: head, S: arm, F: L] as the highest priority, thanks to the best locomotion performance with this combination (indicated by the most number of green color squares in that column). In addition, the combinations of [D: torso, S: arm, F: L] had the second highest priority, followed by [D: head, S: knee, F: L], [D: torso, S: head, F: L] and [D: torso, S: knee, F: L] for body-centric locomotion tasks. These combinations can be recommended for body-centric locomotion in VR.

However, when the participants utilized head-based direction control, they could spend shorter times but result in a larger number of overshoot using head or arm to control speed with CL function. The locomotion performance of the combination [D: head, S: knee, F: CL] was the worst, due to the difficulty of controlling speed using knee with CL function, followed by the combination [D: torso, S: knee, F: CL], [D:

head, S: arm, F: CL], and [D: torso, S: arm, F: CL]. These combinations should be avoided for body-centric locomotion in VR.

5.6 Sample Size for the Experiment

We adopted the G*Power tool [23] to conduct priori power analyses to determine the appropriate sample size with given alpha power and effect size. The effect size set in G*Power was 0.40 (i.e., $\eta_p^2 = 0.14$ in SPSS), considered to be a large effect (the effect sizes obtained in our data analysis are generally greater than this number). With an alpha = 0.05 and power = 0.90, the sample size needed with this effect size is approximately 8 or 10 for the main effects with the number of measurements are 9 or 6 respectively, and 15 for the interaction effects with the number of measurements are 3. Thus, the sample size of 18 should be adequate for the objectives of our study.

5.7 Implications for Design

The findings of our work provide suggestions to VR application designs involving body-centric locomotion tasks. In particular, compared with previous work [4], we considered direction control using body parts to meet the requirements of common locomotion tasked in VR. The combination of body parts for direction control and speed control with various transfer functions could support different purposes of virtual locomotion.

A typical example is VR games, which commonly involve locomotion behavior. For example, if the hands are unavailable to control locomotion interaction [47], users could employ other body-centric locomotion designs, e.g., the use of head for direction control and the use of knee for speed control with L function.

The second example is navigation in a virtual museum. The participants could move slowly or stop movement to look around in VR [18]. Then, the use of torso for direction control and the use of arm for speed control with L function could be adopted for comfortable locomotion experience. In addition, they could speed up irrelevant scenes with CL function, and slow down during approaching the relevant and interesting location with P function.

Based on our results, designers could utilize a combination of body parts with the corresponding transfer functions to provide users comfortable and/or efficient and/or precise interactions.

6 LIMITATIONS AND FUTURE WORK

As a controlled study, we could not cover all body parts and transfer functions. We only examined six combinations using four representative body parts to control speed and direction with three different transfer functions. We collected quantitative and qualitative data with such an experiment setting and gained insights of body-centric locomotion in immersive virtual environments based on data analysis. Our future work will further examine more body parts with more transfer functions. In addition, we will investigate how to seamlessly switch between different body-centric locomotion methods without breaking the sense of presence [27].

Body-centric locomotion with a sitting posture is also of highly practical interest [18], [49]. Head-tilt, arm-swing and torso-lean based virtual locomotion styles are easy to adapt to

a sitting posture. However, for the knee-bend virtual locomotion in a sitting posture, we may need to further take the differences between sitting and standing postures into account. In future work, we would like to compare the performance of body-based locomotion between sitting and standing postures, so as to generalize our conclusions to a greater extent.

In addition, it is worth investigating the body-centric locomotion with a secondary task in VR for future work, looking into the effects of task types on body-centric locomotion. For example, the users may adopt the torso and arm for locomotion, and utilize the gaze for target selection. Such an interaction scenario could be applied to many VR applications (e.g., games, education or assembly design).

7 CONCLUSION

In this study, we systematically investigated the effects of body parts and transfer functions on virtual locomotion in VR. Compared with our previous work [4], we newly included the effects of body parts for direction control to make our conclusion more general and applicable (e.g., VR games or tourism), and adopted more objective and subjective measures to consider more aspects of virtual locomotion, including the number of environmental collision and overshoot, traveled distance, general comfort, and the level of motion sickness. From the experiment, we obtained new combinations of body parts and transfer functions for body-centric locomotion in VR. Our results could provide suggestions for designing body-centric locomotion in VR according to the specific purposes (e.g., precise, efficient, or comfortable experience).

ACKNOWLEDGMENTS

Special thanks to every participant in this experiment and every anonymous reviewer for the comments to improve the paper.

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